

Notes for ‘Sign diagrams’

Important Ideas and Useful Facts:

- (i) **Sign diagram of a function:** The sign diagram of a function indicates
- inputs where the value of the function may be zero or undefined,
 - where the values are positive, where the graph sits above the x -axis, and
 - where the values are negative, where the graph sits below the x -axis.

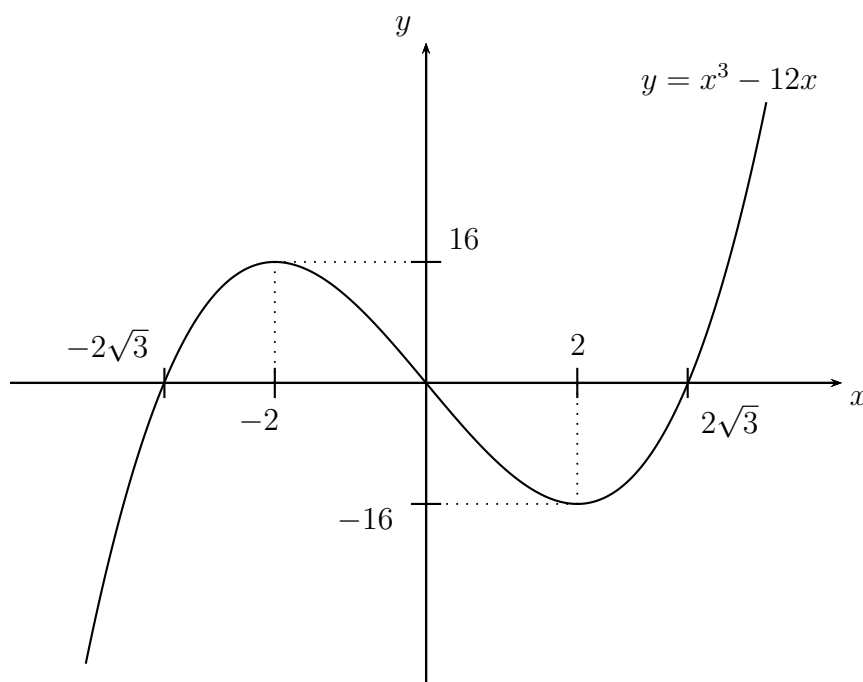
For example, consider the function $y = f(x) = x^3 - 12x$. The rule factorises as

$$f(x) = x^3 - 12x = x(x^2 - 12) = x(x - 2\sqrt{3})(x + 2\sqrt{3}),$$

so that $f(x) = 0$ when $x = 0, \pm 2\sqrt{3}$. By considering the signs of the factors, one sees that $f(x)$ is positive when $x > 2\sqrt{3}$ and when $-2\sqrt{3} < x < 0$. One sees that $f(x)$ is negative when $0 < x < 2\sqrt{3}$ and when $x < -2\sqrt{3}$. This information is summarised in the following sign diagram:

x	$-2\sqrt{3}$	0	$2\sqrt{3}$
$f(x)$	–	0	+
	–	0	+

This information is matched below in the graph of f , where one sees that the graph crosses the x -axis at $x = 0, 2\sqrt{3}$ and $-2\sqrt{3}$, sits above the x -axis over the intervals $(-2\sqrt{3}, 0)$ and $(2\sqrt{3}, \infty)$, and sits below the x -axis over the intervals $(-\infty, -2\sqrt{3})$ and $(0, 2\sqrt{3})$.



(ii) **Sign diagram of the derivative:** The sign diagram of the derivative of a function indicates

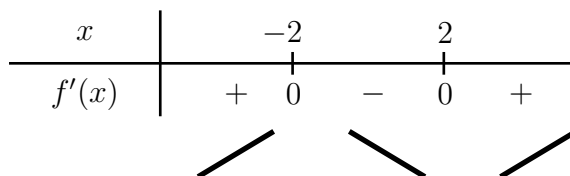
- *critical points* where the value of the derivative may be zero or undefined,
- where the values are positive, where the graph is increasing, and
- where the values are negative, where the graph is decreasing.

We may highlight increasing behaviour visually beneath a plus sign with a symbolic line segment sloping upwards, and decreasing behaviour visually beneath a negative sign with a symbolic line segment sloping downwards.

For example, consider the function $y = f(x) = x^3 - 12x$, whose graph appears above. The derivative is

$$f'(x) = 3x^2 - 12 = 3(x^2 - 4) = 3(x - 2)(x + 2),$$

so that $f'(x) = 0$ when $x = \pm 2$. By considering the signs of the factors, one sees that $f'(x)$ is positive when $x > 2$ and when $x < -2$. One sees that $f'(x)$ is negative when $-2 < x < 2$. This information is summarised in the following sign diagram, with increasing and decreasing behaviour highlighted with sloping up and down symbols:



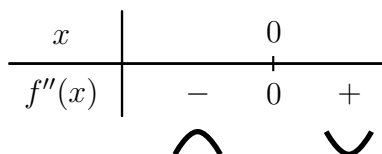
This information is matched in the graph above, where the graph is increasing over the intervals $(-\infty, -2)$ and $(2, \infty)$, and decreasing over the interval $(-2, 2)$.

(iii) **Sign diagram of the second derivative:** The sign diagram of the second derivative indicates

- *critical points* where the value of the second derivative function may be zero or undefined,
- where the values are positive, where the graph is *concave up*, that is, has a ‘bowl-shape up’ appearance, and
- where the values are negative, where the graph is *concave down*, that is, has a ‘bowl-shape down’ appearance.

We may highlight concave up or down behaviour visually beneath any plus signs with a ‘bowl-shape up’ or ‘smiley-face’ symbol, and beneath a negative sign with a ‘bowl-shape down’ or ‘sad-face’ symbol.

For example, consider the function $y = f(x) = x^3 - 12x$, with derivative $f'(x) = 3x^2 - 12$. The second derivative is $f''(x) = 6x$, which has the same sign as x , captured by the following sign diagram, with concave up and down behaviour highlighted with ‘bowl-shape’ up and down symbols:



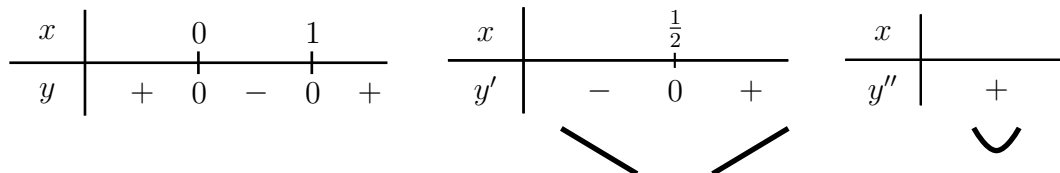
which again matches the behaviour of the curve shown above, which is concave down over the interval $(-\infty, 0)$ and concave up over the interval $(0, \infty)$.

Examples:

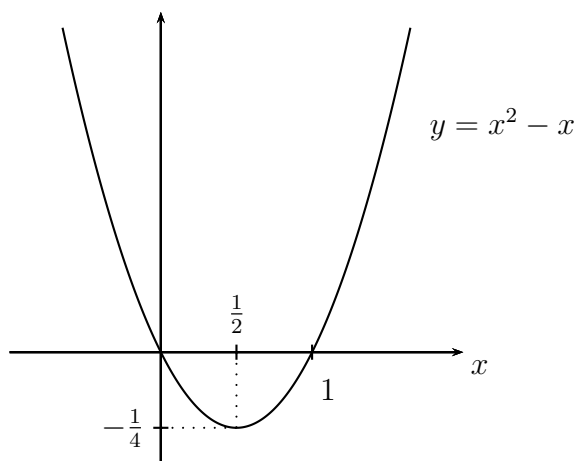
1. Consider the quadratic function

$$y = f(x) = x^2 - x = x(x - 1) .$$

with derivative $y' = 2x - 1$ and second derivative $y'' = 2$, producing the following sign diagrams:



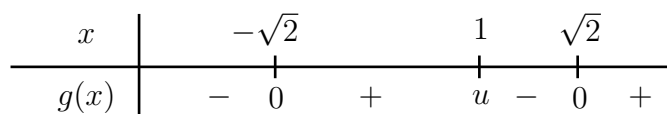
These capture important features of the quadratic. The graph sits above the x -axis for $x < 0$ and $x > 1$ and below the x -axis for $0 < x < 1$. The graph is decreasing over the interval $(-\infty, \frac{1}{2})$ and increasing over the interval $(\frac{1}{2}, \infty)$. The curve is concave up everywhere. Note also the presence of the apex (vertex) of the parabola at the point $(\frac{1}{2}, -\frac{1}{4})$, matching the heuristic symbols associated with the sign diagrams of y' and y'' .



2. Consider the rational function g with rule $g(x) = \frac{x^2 - 2}{x - 1}$, which can be rewritten in different ways, for example,

$$g(x) = \frac{x^2 - 2}{x - 1} = \frac{(x + \sqrt{2})(x - \sqrt{2})}{x - 1} = x + 1 - \frac{1}{x - 1} = x + 1 + \frac{1}{1 - x} .$$

The third and fourth expressions make clear the asymptotic behaviour of the curve, pictured below, which has been discussed previously. Clearly $g(x)$ is undefined at $x = 1$, and $g(x) = 0$ when $x = \pm\sqrt{2}$. By considering signs of the factors in the second expression, we get the following sign diagram for $g(x)$:



This matches the behaviour of the curve, which sits above the x -axis when $x > \sqrt{2}$ and when $-\sqrt{2} < x < 1$, and sits below the x -axis when $x < -\sqrt{2}$ and when $1 < x < \sqrt{2}$.

To get the sign diagram for the derivative, we need to differentiate g and get

$$g'(x) = \frac{d}{dx}(x+1) + \frac{d}{dx}\left(\frac{1}{1-x}\right) = 1 + \frac{d}{dx}\left(\frac{1}{1-x}\right).$$

As practice with one of the limit definitions of the derivative, put $h(x) = \frac{1}{1-x}$. Then, for any $a \neq 1$, we have

$$\begin{aligned} h'(a) &= \lim_{b \rightarrow a} \frac{h(b) - h(a)}{b - a} = \lim_{b \rightarrow a} \frac{\frac{1}{1-b} - \frac{1}{1-a}}{b - a} = \lim_{b \rightarrow a} \frac{1 - a - (1 - b)}{(b - a)(1 - b)(1 - a)} \\ &= \lim_{b \rightarrow a} \frac{b - a}{(b - a)(1 - b)(1 - a)} = \lim_{b \rightarrow a} \frac{1}{(1 - b)(1 - a)} = \frac{1}{(1 - a)^2}. \end{aligned}$$

This shows that $h'(x) = \frac{1}{(1-x)^2}$, so that,

$$g'(x) = 1 + h'(x) = 1 + \frac{1}{(1-x)^2}.$$

Observe that $g'(1)$ is not defined and that $g'(x)$ is positive for all $x \neq 1$, so we get the following sign diagram:

x				
			1	
$g'(x)$		+	u	+

This also matches the behaviour of the curve below, which is increasing both for $x < 1$ and for $x > 1$.

